Voyage-AI — Technical Risk Register & Prioritized Engineering Roadmap

Document Reference: `docs/reports/risks-and-roadmap.pdf` / `docs/reports/risks-and-roadmap.md`
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1. Technical Risk Register

The following risk register categorizes confirmed technical risks, architectural bottlenecks, and operational constraints based on empirical evidence gathered from the codebase and host runtime inspection:

Risk ID	Risk / Issue Description	Supporting Evidence	Likelihood	Impact	Status	Mitigation Implemented	Dependencies	Verification Evid
RSK-01	**Host Port Collision on Port 3306**	Windows service `MySQL80` bound to `0.0.0.0:3306`	High	High	**RESOLVED**	Enforced `3307:3306` port mapping in Docker Compose targeting canonical DB `voyage`; host DB untouched	Docker Compose config	`DatabaseFlyw` verifies 0 conne
RSK-02	**External API Rate Limit (Open-Meteo)**	Relying on public non-commercial endpoint (10k/day)	Medium	Medium	**MONITORED**	In-memory coordinate caching and zero-fabrication fallback	Backend cache	`WeatherApiTe`
RSK-03	**Client Mock Data Divergence & Truncation**	Previously static `mockDestinations.ts` and 100-item ceiling	High	High	**RESOLVED**	Server-side pagination (`page`, `size=12`), search filters, and real Wikimedia ingestion	`DestinationController`	`Destinations` passed)
RSK-04	**Unauthenticated / Cross-User Access**	Private trips exposed without strict user boundaries	High	High	**RESOLVED**	Stateless JWT Bearer filter chain, `UserPrincipal`, and ownership check (`findByUserId`)	Security boundary	`PersistenceT` 401 unauth and
RSK-05	**AI Free-Tier Human Review & PII Leakage**	Gemini free tier policy permits human review	Low	High	**RESOLVED**	Zero user PII transmitted in prompts; prompts constrained to internal POI metadata	AI Adapter	`AiController` verify prompt is
RSK-06	**Administrative Ingestion Endpoint Exposure**	Ingestion endpoints open to non-admin users	High	High	**RESOLVED**	Enforced `@PreAuthorize("hasRole('ADMIN')")` with server-controlled admin email whitelist	Spring Security	`AdminDestina` (3/3 passed)
RSK-07	**Wikimedia Commons Media Licensing Deficit**	Generic attribution violating per-image CC terms	High	High	**RESOLVED**	MediaWiki Action API queries extracting photographer credits and explicit CC licenses	Ingestion engine	`DestinationI` (13/13 passed)
RSK-08	**OSRM Public Routing Server Instability**	Public demo server has no commercial SLA	High	Medium	**PLANNED**	Encapsulated behind `RoutingProvider` interface with Haversine direct distance fallback	Routing adapter	Phase 4 roadma
RSK-09	**OpenStreetMap Tile Usage Policy**	Leaflet client tile requests must adhere to OSM policy	Medium	Low	**PLANNED**	Cache headers and visible attribution configured; CARTO / self-hosted evaluated	Frontend map	Phase 3 roadma

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2. Prioritized Engineering Roadmap

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|                                PRIORITIZED ROADMAP MATRIX                           |
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| Priority Tier | Focus Area                                | Key Deliverables            |
|---------------|-------------------------------------------|-----------------------------|
| **P0 (Critical)**| Security & Persistence Foundation      | Flyway V0-V4, MySQL voyage,  |
|               | (COMPLETED & VERIFIED)                    | JWT auth, roles, profiles    |
| **P1 (High)** | Core Vertical Slice & Real Ingestion      | Paginated destination search,|
|               | (COMPLETED & VERIFIED)                    | Wikipedia/Wikimedia pipeline |
| **P2 (Medium)**| Itinerary, AI & Local Orchestration      | User trip CRUD, Gemini chat, |
|               | (COMPLETED & VERIFIED)                    | start.bat local orchestrator |
| **P3 (Roadmap)**| Spatial Visualization & Cloud Hosting   | Interactive Leaflet maps,    |
|               | (PHASES 3, 4, 7 TARGETS)                  | OSRM routing matrix, CI/CD   |
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3. Engineering Implementation Status & Completed Milestones

3.1 Task 1 (P0): Relational Schema & Persistence Foundation — COMPLETED & VERIFIED

- **Implemented:** Flyway migrations `V0` through `V4` applied to canonical database `voyage` on Docker port 3307. JPA entities (`User`, `Destination`, `TravelListing`, `Trip`, `TripDay`, `ItineraryItem`) with mandatory provenance columns.
- **Verification Evidence:** `DatabaseFlywayIntegrationTest.java` (4/4 passed) and `PersistenceIntegrationTest.java` (3/3 passed).

3.2 Task 2 (P0): Stateless JWT Security & Role Authorization — COMPLETED & VERIFIED

- **Implemented:** Stateless JWT Bearer authentication, user profile management (`/api/v1/users/me`), BCrypt hashing, roles (`ROLE_USER` , `ROLE_ADMIN`), and administrative authorization (`@PreAuthorize("hasRole('ADMIN')")`).
- **Verification Evidence:** `AuthApiTest.java` (10/10 passed), `AdminDestinationControllerTest.java` (3/3 passed), `UserProfileApiTest.java` (2/2 passed).

3.3 Task 3 (P1): Full-Catalog Server Pagination & Search — COMPLETED & VERIFIED

- **Implemented:** `DestinationController` with server-side pagination (`page` , `size=12`), region, style, and budget filter predicates, and pagination headers (`X-Total-Count` , `X-Total-Pages`).
- **Verification Evidence:** `DestinationApiTest.java` (9/9 passed) and `DestinationsDiscoveryTest.java` (5/5 passed).

3.4 Task 4 (P1): Real Wikimedia Ingestion Pipeline — COMPLETED & VERIFIED

- **Implemented:** `DestinationIngestionService` extracting real Wikipedia summaries and MediaWiki Action API photographer credits and CC licenses (`CC-BY-SA 4.0` , `CC0` , `Public Domain`), rendered on cards and modals.
- **Verification Evidence:** `DestinationIngestionServiceTest.java` (13/13 passed) and `DestinationIngestionApiTest.java` (1/1 passed).

3.5 Task 5 (P2): Google Gemini AI Travel Concierge — COMPLETED & VERIFIED

- **Implemented:** `GoogleGeminiService` integrating Google Gemini `gemini-3.5-flash-lite` model with structured error codes, zero user PII transmission, and frontend `AiTravelAssistant` UI.
- **Verification Evidence:** `AiApiTest.java` (3/3 passed) and `AiAssistantBrowserTest.java` (3/3 passed, verified response to "What's capital of India?" in Chromium DOM).

3.6 Task 6 (P2): Local Development Orchestrator (`start.bat`) — COMPLETED & VERIFIED

- **Implemented:** Windows single-command launcher checking MySQL (3307), Backend (8081), Frontend (5173), waiting for readiness, opening default browser, and safely reusing running services without data resets.
- **Verification Evidence:** Validated via Windows `cmd.exe` execution.

3.7 Remaining Roadmap Milestones

- **Task 7 (P3): Interactive Leaflet Maps (Phase 3 Target):** Leaflet.js widget rendering OSM tiles with strict policy compliance and spatial destination plotting.
- **Task 8 (P3): Transit Distance & Routing Matrix (Phase 4 Target):** `RoutingProvider` adapter interface integrating OSRM with automatic Haversine fallback.
- **Task 9 (P3): Production Deployment & Cloud Hosting (Phase 7 Target):** Cloud hosting configuration (Render, Railway, or AWS ECS), SSL/TLS termination, and cloud KMS secrets custody.

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4. Verification Milestones & Quality Gates

- **Milestone 1 (Persistence & Security Gate):** PASSED — Flyway migrations V0-V4 applied cleanly; JPA repositories verified; all `/api/v1/trips/**` endpoints reject unauthenticated requests with HTTP 401.
- **Milestone 2 (Dynamic Vertical Slice Gate):** PASSED — React client renders paginated destinations fetched from `/api/v1/destinations` with real Wikimedia CC licensing.
- **Milestone 3 (AI & Meteorological Gate):** PASSED — Gemini AI adapter and Open-Meteo weather service verified live without mocks.
- **Milestone 4 (Full System Quality Gate):** PASSED — 128 automated tests passing (83 backend, 11 frontend, 34 Playwright) with 100% pass rate. Single canonical database `voyage` enforced on port 3307. Single-command orchestrator `start.bat` verified.